

Grundlagen der künstlichen Intelligenz

Los Del DGIIM, losdeldgiim.github.io

Doble Grado en Ingeniería Informática y Matemáticas
Universidad de Granada

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1. Introduction

Artificial Intelligence (AI) is a field that is rapidly evolving and has the potential to revolutionize various industries. Before delving into the technical aspects of AI, it is important to understand the concept of intelligence and how it can be measured.

1.1. Intelligence Tests

Scientists have tried to decide what differentiates AI from computer algorithms; i.e. what is the essence of intelligence? This question has been debated for decades, and there are different tests that have been proposed to determine whether a machine can be considered intelligent or not. The importance of these tests is shown in the Chinese Room thought experiment.

- Chinese Room Experiment:

The Chinese Room thought experiment involves a person who does not understand Chinese sitting in a room with a set of instructions for how to respond to Chinese characters. The person receives Chinese characters as input and uses the instructions to produce appropriate responses, which are then sent back out of the room.

From the outside, it may appear that the person in the room understands Chinese, but in reality, they are simply following a set of rules without any comprehension of the language. This thought experiment raises questions about whether a machine can truly understand language or if it is simply following a set of instructions.

Some of the most well-known tests include:

- Turing Test:

In the Turing Test, a human evaluator interacts with both a machine and a human without knowing which is which. If the evaluator cannot reliably distinguish between the machine and the human, then the machine is said to have passed the Turing Test and is considered to be intelligent.

It only tests for the ability to mimic human behavior, not for true understanding or consciousness.

- Winograd Test:

The test involves a series of sentences known as “Winograd Schemas”. They include ambiguous pronouns that require common sense and contextual understanding to resolve.

This test is not about factual knowledge but rather about understanding context.

- Lovelace Test:

The Lovelace Test is designed to evaluate a machine's ability to create something original and surprising.

This test is extremely difficult for machines to pass, as they often just combine existing ideas rather than creating something truly new.

- Metzinger Test:

The Metzinger Test is designed to evaluate a machine's ability to have a subjective experience, a sense of “self, I”, and consciousness. There is no definitive method for conducting the test in practice, since subjective experience cannot be observed from the outside.

1.2. Initial Definitions

From more general concepts to more specific ones, where each one builds upon the previous, some important terms in the field of AI include:

- Data Science: Field that turns data into knowledge and decisions.
- Artificial Intelligence: Field that aims to create machines that can perform tasks that typically require human intelligence.
- Machine Learning: Subfield of AI that focuses on training machines to learn from data and improve their performance over time without being explicitly programmed. There are three main types of machine learning:
 - Supervised Learning: The machine is trained on labeled data, where the correct output is provided for each input.
 - Unsupervised Learning: The machine is trained on unlabeled data, where it must find patterns and relationships on its own.
 - Reinforcement Learning: The machine learns by interacting with an environment and receiving feedback in the form of rewards or penalties based on its actions.
- Neural Networks: A type of (usually supervised) machine learning model inspired by the structure and function of the human brain. They consist of layers of interconnected nodes (neurons) that process and transmit information.
- Deep Learning: Neural networks with multiple layers. Deep learning has been particularly successful in tasks such as image recognition, natural language processing, and speech recognition.

2. Data Science

Data Science is a multidisciplinary field that combines statistical analysis, machine learning, and domain expertise to extract insights and knowledge from data. Data is the raw material that fuels the development of AI systems, and understanding how to work with data is crucial for anyone interested in the field of AI. Two main people work on it:

- Data Engineer: Responsible for building and maintaining the infrastructure that allows for the collection, storage, and processing of data.
- Data Scientist: Responsible for analyzing and interpreting data to extract insights and knowledge.

Regarding data collection, there are some aspects that must be taken into account:

- Data Quality: The quality of the data is crucial for the success of any AI project.
- Data Integrity: Ensuring that the data is accurate and reliable.
- Data Consistency: Ensuring that the data is consistent across different sources and formats.

Once the data has been collected, it must be preprocessed before it can be used for training AI models. This involves:

- Data Validation: Checking the data for errors and inconsistencies. Depending on the source and type of data, some rules can be applied to validate it.
- Data Cleaning: Removing or correcting any errors or inconsistencies in the data.
- Data Transformation: Converting the data into a format that can be used for training AI models.

Data preprocessing also involves data scaling, which is the process of normalizing or standardizing the data to ensure that it is on the same scale. This is important because many AI algorithms are sensitive to the scale of the data and can perform poorly if the data is not scaled properly. Some common scalers include:

- **Binarizer**: Converts categorical data into binary data. A threshold is set, and values above the threshold are converted to 1, while values below the threshold are converted to 0.

$$X' = \begin{cases} 0 & \text{if } X \leq \text{threshold} \\ 1 & \text{if } X > \text{threshold} \end{cases}$$

- **MaxAbsScaler**: Scales the data to the range $[-1, 1]$ by dividing each value by the maximum absolute value in the data.

$$X' = \frac{X}{\max(|X|)}$$

- **MinMaxScaler**: Scales the data to a specified range, let's say $[\text{mín}, \text{máx}]$. Firstly the data is scaled to the range $[0, 1]$ (X_{std}), and then it is scaled to the specified range.

$$X_{\text{std}} = \frac{X - \text{mín}(X)}{\text{máx}(X) - \text{mín}(X)} \quad X' = X_{\text{std}} \cdot (\text{máx} - \text{mín}) + \text{mín}$$

- **Normalizer**: Scales the data to have a unit norm. This is done by dividing each value by the norm of the data. Usually, the L2 norm is used.

$$X' = \frac{X}{\|X\|}$$

- **PowerTransformer**: Applies a power transformation to the data to make it more Gaussian-like. This is useful for data that is not normally distributed.
- **RobustScaler**: Scales the data using statistics that are robust to outliers. This is useful for data that contains outliers.

$$X' = \frac{X - \text{median}(X)}{\text{IQR}(X)}$$

where $\text{IQR}(X)$ is the interquartile range of X .

- **StandardScaler**: Scales the data to have a mean of 0 and a standard deviation of 1.

$$X' = \frac{X - \mu}{\sigma}$$

where μ is the mean of X and σ is the standard deviation of X .

One important aspect of data preprocessing is *encoding* the data. This is the process of converting categorical data into numerical data that can be used for training AI models. There are several techniques for encoding data, such as:

- **Ordinal Encoding**: Assigning a unique integer to each category in the data. The order of the integers is important, as it implies a relationship between the categories.

$$\{\text{small, medium, large}\} \rightarrow \{0, 1, 2\}$$

- Label Encoding: Similar to ordinal encoding, but it does not imply any relationship between the categories. Each category is assigned a unique integer, but the order of the integers does not matter.

$$\{\text{red, green, blue}\} \rightarrow \{0, 1, 2\}$$

- One-Hot Encoding: Creating a binary column for each category in the data.

$$\{\text{red, green, blue}\} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Dummy Encoding (Leave out the last): Similar to one-hot encoding, but it leaves out the last category to avoid multicollinearity.

$$\{\text{red, green, blue}\} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

- Absolute Frequency Encoding: Assigning a unique integer to each category based on the absolute frequency of the category in the data.

$$\begin{aligned} &\{\text{red, red, red, green, blue, blue}\} \\ &\{\text{red, green, blue}\} \rightarrow \{3, 1, 2\} \end{aligned}$$

- Relative Frequency Encoding: Assigning a unique integer to each category based on the relative frequency of the category in the data.

$$\begin{aligned} &\{\text{red, red, red, green, blue, blue}\} \\ &\{\text{red, green, blue}\} \rightarrow \{3/6, 1/6, 2/6\} \end{aligned}$$

- Feature Hash Encoding: Hashing the categories into a fixed number of columns. This is useful for data with a large number of categories.

Once the data has been preprocessed, they can be used. There are two main destinations for the data:

- Data Visualization: The process of creating visual representations of data to help understand and communicate insights. This is used by people.
- Data Modeling: The process of using data to train AI models. This is used by machines.

3. Search Algorithms

In our current world, plenty of algorithms are used to search for information. From the most basic ones, such as sorting algorithms, to the most complex ones, such as search engines. In this chapter, we will explore some of the most common search algorithms and their applications.

3.1. Graph Theory

Search algorithms are often used in graphs, as they provide a way to represent complex relationships between data. A graph $G = (V, E)$ is defined as a pair of sets, where V is called the set of vertices and E is called the set of edges. The vertices represent the data points, while the edges represent the relationships between them. There are different types of graphs, such as directed and undirected graphs, weighted and unweighted graphs, or cyclic and acyclic graphs. Each type of graph has its own properties and algorithms that can be applied to it. An example graph is shown in Figure 3.1.

A graph is usually represented as an *adjacency matrix*. Where the rows represent the origin vertices, the columns represent the destination vertices¹, and the values a_{ij} represent:

- If the graph is unweighted, a_{ij} represents the number of edges from vertex i to vertex j .
- If the graph is weighted, a_{ij} represents the weight of the edge from vertex i to vertex j . If there is no edge, a_{ij} is usually set to infinity or an invalid value.

For example, the adjacency matrix for the graph in Figure 3.1 is shown in Equa-

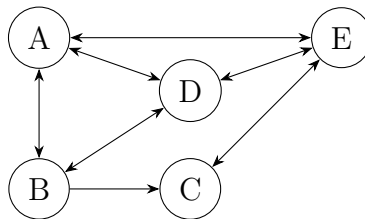


Figure 3.1: Example of a directed graph.

¹This could also be the opposite: the origin vertices could be the columns and the destination vertices, the rows.

tion 3.1.

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix} \quad (3.1)$$

This matrix is also called “next-neighbor” matrix, as the neighboring vertices of a vertex i can be found by looking at the i -th row of the matrix. The “next-next-neighbor” matrix, on the other hand, is obtained by multiplying the adjacency matrix by itself. This matrix can be used to find the vertices that are two steps away from a given vertex.

Lastly, the *transition matrix* is obtained by normalizing the adjacency matrix. This is done by dividing each entry a_{ij} by the sum of the entries in the i -th row. The transition matrix can be used to find the probability of transitioning from one vertex to another in a random decision process.

3.2. Sorting Algorithms

Sorting algorithms are a fundamental part of computer science, as they allow us to organize data in a specific order. In addition, they are used in many search algorithms, as they can help to reduce the search space and improve the efficiency of the search.

3.2.1. Insertion Sort

This algorithm works by dividing the array into a sorted and an unsorted part. The sorted part is initially empty, and the unsorted part contains all the elements.

1. Start with the first element of the unsorted part and compare it with the last element of the sorted part.
2. Move all the elements of the sorted part that are greater than the current element to the right.
3. Insert the current element into its correct position in the sorted part.
4. Repeat the process for all the elements in the unsorted part until it is empty.

It has a time complexity of $O(n^2)$.

3.2.2. Bubble Sort

This algorithm works by repeatedly swapping adjacent elements if they are in the wrong order. The process is repeated until the array is sorted.

1. Compare each pair of adjacent elements in the array.
2. If the elements are in the wrong order, swap them.
3. Repeat the process for all the elements in the array until it is sorted.

It has a time complexity of $O(n^2)$.

3.2.3. Shaker Sort

This algorithm is a variation of Bubble Sort that sorts in both directions on each pass through the list. It works by comparing adjacent elements and swapping them if they are in the wrong order, similar to Bubble Sort. However, it also compares elements in the opposite direction, which can help to reduce the number of passes needed to sort the array.

1. Start at the beginning of the array and compare each pair of adjacent elements.
2. If the elements are in the wrong order, swap them.
3. After reaching the end of the array, reverse the direction and compare each pair of adjacent elements again.
4. Repeat the process until the array is sorted.

It has a time complexity of $O(n^2)$, but with lower constant factors than Bubble Sort.

3.2.4. Quick Sort

This divide-and-conquer algorithm works by selecting a pivot element from the array and partitioning the other elements into two sub-arrays, according to whether they are less than or greater than the pivot. The sub-arrays are then sorted recursively. This process continues until the base case of an empty array or an array with a single element is reached.

1. Choose a pivot element from the array.
2. Partition the array into two sub-arrays: one with elements less than the pivot and one with elements greater than the pivot.
3. At this point, the pivot is in its final position.
4. Recursively apply the above steps to the sub-arrays until the base case is reached.

The efficiency of Quick Sort depends on the choice of the pivot. The complexity depends on the partitioning of the array. If the pivot divides the array into two equal halves, the time complexity is $O(n \log n)$. However, if the pivot is consistently the smallest or largest element, the time complexity degrades to $O(n^2)$.

3.2.5. Topological Sorting: Kahn's Algorithm

This algorithm is used to sort the vertices of a directed acyclic graph (DAG) in a linear order. If there is a directed edge from vertex u to vertex v , then u comes before v in the ordering. It is used in several applications, such as scheduling tasks, resolving dependencies, and determining the order of compilation in programming languages.

3.3. Uninformed Search

3.4. Informed Search

3.4.1. Dijkstra Search

3.5. Local Search

3.6. Modern utilities

3.6.1. Information Retrieval

3.6.2. Search Engines

3.6.3. PageRank

4. Übungen

4.0. Introduction

The Python part of this exercise is available in [this Jupyter Notebook](#).

Intelligence Tests

Ejercicio 1. Turing Test vs. Lovelace Test: Which test focuses on creativity and which on simulating conversation? How do these two tests work?

The Turing Test focuses on simulating conversation. It works by having a human evaluator engage in a natural language conversation with both a machine and a human without knowing which is which. If the evaluator cannot reliably distinguish the machine from the human, the machine is considered to have passed the Turing Test.

The Lovelace Test, on the other hand, focuses on creativity. It requires an AI to create something original (like a piece of art, music, or a story) that cannot be attributed to any human creator. The AI must be able to explain how it created the work and why it is original, demonstrating a level of creativity that goes beyond mere imitation.

Ejercicio 2. Why do many simple chatbots fail the Winograd Schema?

Many simple chatbots fail the Winograd Schema because they lack the necessary contextual understanding and common sense reasoning required to correctly interpret the ambiguous sentences used in the test. The Winograd Schema consists of sentences that have a pronoun with an ambiguous antecedent, and the correct interpretation depends on understanding the context and the relationships between the entities involved. Simple chatbots often rely on pattern matching or statistical methods rather than true comprehension, which leads to incorrect answers when faced with such nuanced language challenges.

Ejercicio 3. What is the main idea behind the Metzinger test for checking if an AI is self-aware, in simple terms? What is one thing that might make this test difficult to use for all types of AI?

The main idea behind the Metzinger test is to check if an AI can understand and describe its own internal processes and experiences, which would indicate self-awareness. One difficulty with this test is that it relies on the AI's ability to communicate its thoughts and feelings, which may not be possible for all types of AI,

especially those that do not have a natural language processing capability or a way to express their internal states effectively. Therefore, there is no universally applicable method to determine if an AI is self-aware, as it may depend on the specific design and capabilities of the AI in question.

4.1. Data Processing / Cleaning

The Python part of this exercise is available in [this Jupyter Notebook](#).

Data Filtering and Transforming

Ejercicio 4.1.1. Imagine you have a list of all the students in a school. You only want to look at the students in the 5th grade (data filtering). Then, you want to calculate the average age of those 5th graders (data transforming). Explain why you need both steps.

We first need to filter the data to focus only on the students in the 5th grade, as we are specifically interested in that group. This step allows us to narrow down our dataset to the relevant subset of students. After filtering, we need to transform the data by calculating the average age of those 5th graders. This transformation step is necessary to derive meaningful insights from the filtered data, as it provides us with a summary statistic (the average age) that can help us understand the characteristics of the 5th-grade students. Without filtering, we would be calculating the average age of all students, which would not give us the specific information we are looking for about the 5th graders.

Ejercicio 4.1.2. You have a list of toys that were sold in a store, including the toy's name and the price it sold for.

1. Give an example of a question you could answer by only looking at toys that cost more than 20 (data filtering).

A question that could be answered by data filtering would be: "Which toys that cost more than 20 euros were sold in the last month?" (Here, we filter the data by price greater than 20 euros and potentially by the sales date).

2. Give an example of something you could calculate or change about the toy prices (data transforming).

An example of data transformation would be calculating the average selling price of all toys or converting all prices from euros to another currency (e.g., US dollars) based on a specific exchange rate. Another transformation could be creating a new column indicating whether a toy is classified as "cheap" (e.g., < 15 euros), "mid-priced" (e.g., 15 euros – 30 euros), or "expensive" (> 30 euros) based on the selling price.

Data Understanding

Ejercicio 4.1.3. What is the difference between quantitative (numerical) and qualitative (categorical) data?

Quantitative data refers to numerical values that can be measured and quantified, such as height, weight, or age. Qualitative data, on the other hand, refers to categorical values that describe characteristics or attributes, such as color or gender. Quantitative data can be analyzed using mathematical operations, while qualitative data is typically analyzed using descriptive statistics or categorization.

Ejercicio 4.1.4. What does the abbreviation **NaN** mean in a dataset?

NaN stands for “Not a Number” and is used in datasets to represent missing or undefined values. It is also used to indicate that a value is not a valid number, such as the result of an undefined mathematical operation (e.g., division by zero). In data analysis, **NaN** values need to be handled appropriately, either by filling them with a specific value or dropping them.

Ejercicio 4.1.5. Briefly explain the difference between the mean and the median. Which value is more sensitive to outliers?

The mean is the average of a set of numbers, calculated by summing all the values and dividing by the count of values. The median is the middle value in a sorted list of numbers. The mean is more sensitive to outliers because it can be significantly affected by extremely high or low values, while the median is less affected as it only considers the middle value(s) and not the magnitude of all values.

Ejercicio 4.1.6. Calculate the quartiles for the following dataset: [10, 9, 1, 4, 3, 2, 5, 6, 6, 6, 3, 9].

First, we need to sort the dataset:

[1, 2, 3, 3, 4, 5, 6, 6, 6, 9, 9, 10]

The quartiles are then:

$$Q2 = \frac{6 + 5}{2} = 5,5$$
$$Q1 = \frac{3 + 3}{2} = 3$$
$$Q3 = \frac{6 + 9}{2} = 7,5$$

Data Cleaning

Ejercicio 4.1.7. Name two reasons why data is often dirty in the real world.

Data can be dirty due to:

- Human error during data entry, which can lead to typos, incorrect values, or missing data.
- Inconsistent data formats or structures, such as different date formats or varying units of measurement across datasets.

Ejercicio 4.1.8. What happens when you apply the `dropna()` function to a DataFrame?

The `dropna()` function removes any rows (or columns, if specified) that contain missing values (**NaN**) from the DataFrame. This is a common data cleaning step to ensure that the dataset does not contain incomplete records that could affect analysis or modeling.

Ejercicio 4.1.9. Why do you scale (normalize) numerical data before feeding it into a machine learning model?

Scaling (normalizing) numerical data is important because many machine learning algorithms are sensitive to the scale of the input features. If the features have different ranges, the algorithm may give more weight to those with larger values, leading to biased results. Scaling helps to ensure that all features contribute equally to the model's learning process and can improve the convergence speed and performance of the model.

Ejercicio 4.1.10. Name two possible encodings and explain them using an example.

They are explained in the page 8.

4.2. PONER

The Python part of this exercise is available in [this Jupyter Notebook](#).

Kahn's Algorithm (Topological Sorting)

You are responsible for the course planning at a university. For a specific module, there are several lectures that partially build upon each other. Your task is to find an order in which all lectures can be attended without attending a lecture before one of its prerequisites.

Lectures (Nodes): Math 1, Math 2, Computer Science 1, Computer Science 2, Databases, Algorithms, Software Development

Dependencies (directed edges “must be attended before”): ■ Math 1 → Math 2

- Computer Science 1 → Computer Science 2
- Computer Science 1 → Algorithms
- Math 2 → Algorithms
- Computer Science 2 → Databases
- Algorithms → Software Development

Ejercicio 4.2.1. Create the graph. Represent the directed graph as an adjacency list, and for each node, note the in-degree (number of incoming edges).

Ejercicio 4.2.2. Apply Kahn's Algorithm. Perform Kahn's algorithm manually (or implement it in a programming language of your choice) to find a topological sorting of the lectures. Document each step:

1. Start with all nodes whose in-degree = 0.
2. Remove one of these nodes, add it to the order, and reduce the in-degrees of its neighbors.
3. Repeat until all nodes are processed.

Ejercicio 4.2.3. Interpret the result. State the calculated order of the lectures, and determine if this order is unique. Justify your answer.